Problem: **Yuna ate 1/4 of a pizza and her brother ate 2/5 of it. How much did they eat together?**

Learner persona: **Elementary, grade 3-4 (age 8-10)**

Step-DPO (baseline)

Step 1: First, find how much pizza Yuna ate. She ate 1/4 of the pizza.

Step 2: Next, her brother ate 2/5 of the pizza.

Step 3: To find the total, we add the fractions 1/4 and 2/5.

Step 4: To add these fractions, we need a common denominator. The least common denominator for 4 and 5 is 20.

Step 5: Convert 1/4 to 5/20 and 2/5 to 8/20. Add them to get 13/20.

Final answer: 13/20

Jumps to the forbidden term 'common denominator' - no concrete grounding for the child.

SLC-StepDPO (Ours)

Step 1: First, find how much Yuna ate. To make it easier, think of the pizza as 4 equal parts. Yuna took 1 part, so she ate 1 out of 4 parts.

Step 2: Now her brother. Imagine the pizza as 5 equal parts. He took 2 parts, so 2 out of 5 parts.

Step 3: To add them, think of the pizza as having 20 equal parts. (a common multiple of 4 and 5). Yuna ate 5 parts and her brother ate 8 parts, so 5 + 8 = 13 parts.

Step 4: The whole pizza is 20 parts, so together they ate 13 out of 20.

Final answer: 13/20

Grounds every step in 'equal parts' of a pizza - the level-appropriate representation.

Figure. Same fraction problem for an elementary learner (greedy decoding, real outputs). Red = forbidden formal term; Green = level-appropriate 'equal parts' framing. Both reach 13/20.